



Loops: Code That Repeats

Quick facilitation notes & answer key

Goal

Students learn that a repeat loop runs the blocks inside it a set number of times. They count the repeats, change the number, and write their own.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app (Scratch-style) to project for the worked example; the printed worksheet and a pencil per student.

Run it (about 30 min)

1. Hook: have the class clap 4 times. Say it as a loop — "repeat 4: clap."
2. Show the repeat-4 square loop. Count the runs out loud together.
3. Students do Exercises 1–3 on their own or in pairs.
4. Share answers. Land the big idea: the repeat number = how many times.

Answer key (checked by the run-gate)

Ex 1 — Bit moves 4 times and turns 4 times.

Ex 2 — Bit moves 6 times. That is MORE than before.

Ex 3 — repeat 5.

Watch for & differentiate

Common slip: counting the blocks once instead of once per repeat.

Support: read "repeat 6" as "do it 6 times" and tap each run on fingers.

Extend: ask how many TOTAL moves if the loop had 2 move blocks and repeat 3.